EPSRC Undergraduate Vacation Scholarship Report

Project name:
Extending an agent-based model of chemotaxis in *Drosophila* larvae

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Approximate project dates:
Monday 20th June 2016 to Friday 26th August 2016 (10 weeks)

Location:
Informatics Forum, School of Informatics, University of Edinburgh

1 Non-Academic Report

1.1 Non-technical summary

The purpose of the project was to improve an existing computer model [1] of the way *Drosophila* larvae (maggots) move towards odours they like (this kind of movement is called chemotaxis). The model describes how the direction a larva is moving is determined by what it is sensing (eg. through smell). The agent (simulated larva) is represented as simply as possible for this task, as a single point in 2D space (its position) with an associated direction (its orientation). In the model, time is discrete (progresses in chunks) rather than continuous. Senses are modelled by feeding the agent a 'sense value' at the start of each time step, with the current sense value depending on the agent's position in its environment. Movement is modelled by changing the agent's position and orientation each time step. The agent's environment is modelled as closely as possible to that of a Petri dish in which experiments on real-life larvae are performed, with the sense values in the virtual environment taken from actual data of the odour concentrations during such an experiment.

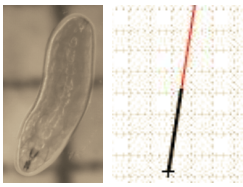

Figure 1: Larva vs single-point agent.

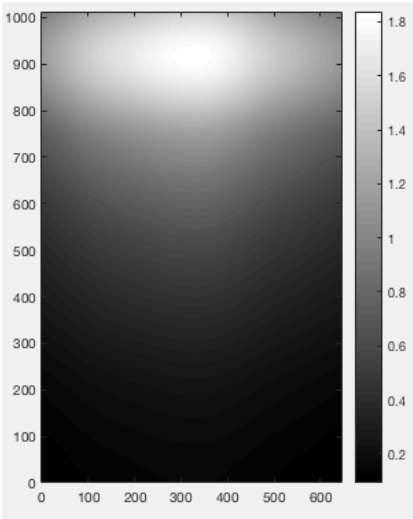

Figure 2: The virtual environment; shading indicates sense (odour) values.

A limitation of this single-point model is that it cannot capture a feature of larval movement thought to be important to chemotaxis. This is the 'head cast', whereby the larva stops its forward movement and moves its head from side to side, bending its body in the process. The first task of this project was to change the code of the model (written in a programming language called MATLAB) so that the agent is represented by three connected points in space rather than a single point. This made the main task of the project possible, which was to change the code of the model so that the agent performs this 'head cast' behaviour. As well as completing the main task, a further achievement was writing code that can create videos displaying the movement of agents (single-point or three-point) through the virtual environment. Example videos can be accessed from[1][2][3][4].

The final task of the project was to write code for virtual experiments that can be performed on the agents in order to generate meaningful statistics (metrics) that can be analysed to try and answer a scientific question about larval 'head casts' - how exactly do head casts (or 'stopping behaviour' as it is also known) effect chemotaxis performance?

[1] single-point agent: `https://youtu.be/2OHcIFFPM7M`
[2] three-point agent: `https://youtu.be/s6yhRlGXD8w`
[3] three-point agent with random stops: `https://youtu.be/MUk4yVOGWXA`
[4] three-point agent with sensory-triggered stops: `https://youtu.be/i_ZX-w1Y3-w`

The analysis code was completed successfully, but only a limited number of experiments were possible in the time remaining. Nevertheless some interesting conclusions could be drawn: head casts increase the straightness of the path the agent takes towards the source of the odour, but too many head casts will slow the agent down, and reduce the benefit of having this straighter path - there may however be an optimal probability of executing head casts that will ultimately get the agent to the odour source quicker than had it not carried out any head casts at all. Further, if the agent executes head casts only when it senses it is moving away from the odour (sensory-based stops), this will allow the agent to stay closer to the odour source once it has been reached.

1.2 Breakdown of time spent

Weeks 1-2 Preliminary work - I familiarised myself with the code for the model, and some of the background theoretical work.

Weeks 3-4 Preliminary work - I created the video function for displaying the agent's movements in detail.

Weeks 5-6 Main work - I extended the model to include multiple points.

Week 7 Main work - I extended the model to include stopping behaviour.

Weeks 8-9 Experiments - I wrote the code for running the experiments, calculating each of the metrics, and displaying the resulting graphs.

Week 10 Report - I ran the experiments, generated the final results, and wrote the report.

1.3 Personal statement

Carrying out the project has helped to develop and improve my skills and experience in a number of areas:

- The project has given me the opportunity to develop several transferable skills, including proposal writing, project planning and coordination, communicating with academic staff, reviewing academic literature, academic writing and analysis, and working within a research team.
- The project has given me valuable insight into how research is performed in practice, and what kind of every-day duties and tasks are carried out by postgraduate students and research assistants.
- The project has been excellent preparation for my upcoming undergraduate project, giving me familiarity with the Informatics Forum robotics lab, and establishing working relationships with my supervisor and other students and researchers within the lab.
- The project has made me more familiar with the programming language MATLAB, and given me general programming and software engineering experience. The project has also given me basic exposure to some of the other disciplines important to the interdisciplinary field of biorobotics, including biology and neuroscience.

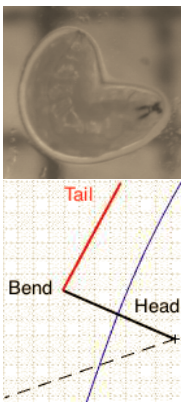

Figure 3: Head cast for larva vs three-point agent.

Overall, I am glad to have done the project, and believe the experience and insight I have gained will be of benefit to my future studies and beyond.

1.4 Financial acknowledgements and adequacy

I would like to thank EPSRC for the financial support provided throughout the duration of the project. This was adequate, and was used to pay for rent and living expenses over the summer period.

2 Academic Report

2.1 Introduction and aims

One approach to understanding how animal brains control behaviour is to build simulations in which hypothesised algorithms for the underlying neurological mechanisms can be tested. By focusing on particular behaviours in relatively 'simple' animals, such as the larvae of *Drosophila*, (which have only a few thousand neurons), a complete model of the natural system can potentially be built.

This project focuses on chemotaxis behaviour in *Drosophila* larvae. There are conflicting explanations for this behaviour in the literature. One explanation states that the larvae "actively sample" chemical odours (similar to sniffing in animals) with distinct control mechanisms triggering specific actions (for example, stop or turn) based on the sensory information gathered [2]. Whilst a model implementing this hypothesis has produced behavioural data similar to larvae, the complexity of the model is high, with three separate control mechanisms and multiple behavioural states [3].

A more recent explanation states that larvae move and turn via "direct sensory modulation of continuous lateral oscillations of the anterior body" [1]. In this simpler hypothesis, no active decision making is required; instead movement is driven by a continuous rhythm within the neural circuitry of the animal, and sensory information controls how this rhythm translates into movement. This is achieved using the most minimal larval representation possible - a single point in space coupled with an associated direction of travel. The behaviour and simplicity of [1] compare favourably to [3]. However, [1] is currently unable to produce larval 'stopping behaviour', whereby the larvae stop their forward motion and display large lateral head sweeps. This behaviour is thought to be important to chemotaxis, and is the subject of this investigation.

The aims of this investigation are as follows:

1. Extend the model from [1] so that the larval agent is represented by three connected points in space (representing the head, bend-point and tail of the organism), rather than the single oriented point of the current model.
2. Extend the model further to produce larval 'stopping behaviour'.
3. Perform experiments to compare the behaviour of the three-point model with and without stopping behaviour. Suitable behavioural metrics must be chosen for a meaningful comparison.
4. In light of the results, discuss the effect of stopping behaviour on chemotaxis.

2.2 Methods

Details of the single-point model are given by [1]. The movement algorithm for the agent is summarised here. At each time step, which represents 1 second, the agent does the following:

1. Sense the difference ∇s between the current sensory (odour) value, and the sensory value at the previous time step.
2. Rotate on the spot by angle α_i, calculated using $\alpha_i = bound(\alpha_b + \nabla s * gain) * oscillation_direction$, where α_b is the baseline turn angle, usually 10°, the *gain* is a model parameter representing how attracted the agent is to the stimulus ($gain < 0$ for a positive attraction),

and the oscillation direction ($left = 1, right = -1$) alternates every time step. The *bound* function forces α_i within the range $[0°, 180°]$ in the current *oscillation_direction* so that the agent never turns in the opposite direction, and never turns more than 180° in a single time step. The heading of the agent at time-step i, θ_i, is given by $\theta_i = \theta_{i-1} + \alpha_i$.

3. Move forward in the new heading θ_i by a set amount (representing 1mm).

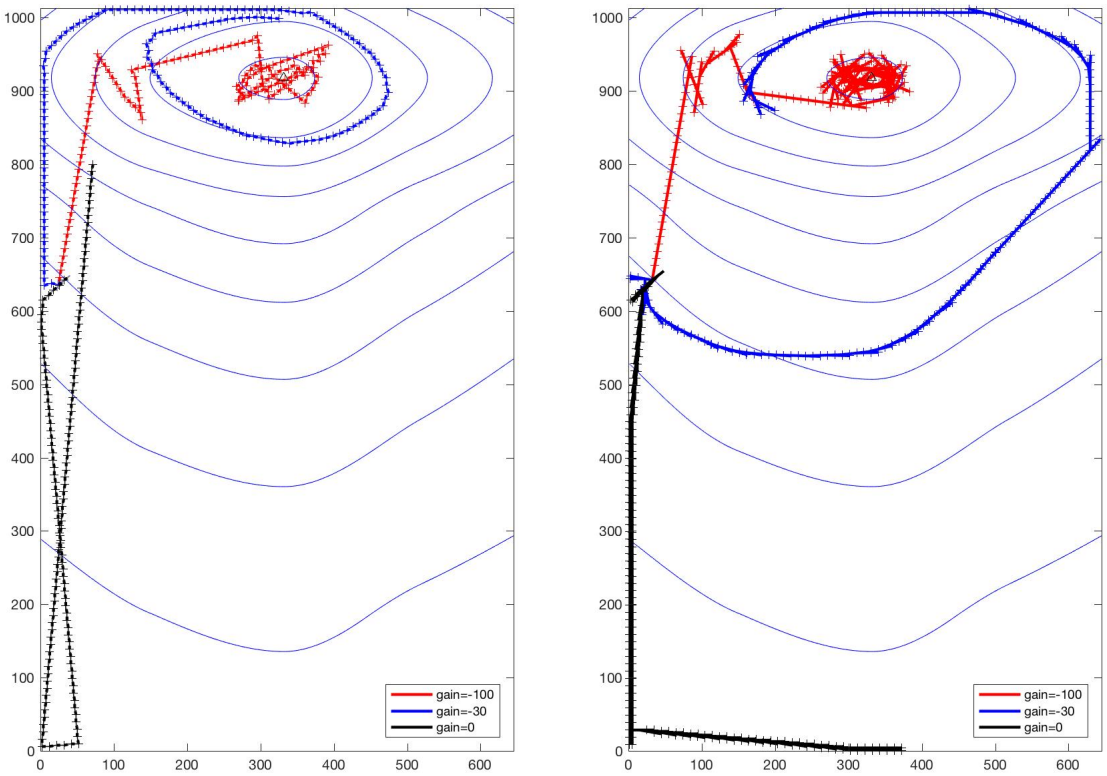

Figure 4: Effect of gain change on the single-point (left) and three-point (right) agent paths.

In order to introduce stopping behaviour into the model, it was required that the the larva is represented using multiple points rather than a single point. Specifically, the agent must have a head segment in order to perform 'head casts' where the rest of the agent - the tail segment - remains stationary. The point connecting the two segments is modelled explicitly, and is called the 'bend point'. The multi-point model allows an arbitrary number of equidistant points to be used with any point designated as the bend point (except the head or tail point). Throughout this investigation however, the minimal three equidistant points are used to represent the larva, with the bend point as the midpoint of the head-tail line. Section 2.4 discusses this decision further. The length of the full agent represents 4mm (2 * 2mm segments).

Once the agent was changed to consist of three points, the next step was to separate the turn and forward movement portions of the movement algorithm, so that it is possible for the agent to turn its head without moving forward (the stopping behaviour). In the single-point model, the point rotates on the spot, and so only the new heading θ_i must be calculated. In

the three-point model however, the head segment pivots from the bend point such that the bend-head line faces the new heading θ_i. This means the new position of the head point must also be calculated. The forward movement of the agent also becomes more complicated - the head and bend points progress in the heading θ_i as the single-point agent does, but the tail point does not. Instead the tail point moves such that the head-tail line gradually straightens out as the agent moves forward without turning (inspired by [3]). The calculation used was that the angle of the bend-tail line moves towards $-\theta_i$ at the rate of 180° in the time it takes the agent to move forward its full body length (4 seconds).

As can be seen in Figure 4, the change to three-points affects the path produced by the agent. It was further found that the order of operations - sense, turn, move forward (STM) vs sense, move forward, turn (SMT) also changes the paths produced. The reasons for these changes are discussed in Appendix A. It is enough now to say that the SMT order was chosen throughout this investigation. In changing the model to three-points, physical constraints had to be introduced to prevent the agent bending-over itself, and to deal with what happens when the agent collides with a wall. These issues are detailed in Appendix B.

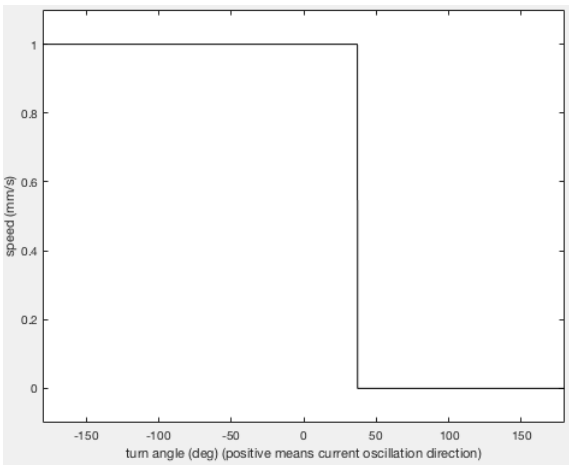

Figure 5: Depiction of the `piecewise` speed function.

The stopping behaviour itself was implemented by introducing a variable *speed* for the amount the agent moves forward each time step. A stop occurs when the *speed* is 0 for a particular time-step. With stops now possible in the model, the issue of when a stop occurs was addressed. Rather than modelling stops as discrete events, the approach was taken such that the current signal change ∇s and *gain* modulate *speed* in a similar way to how they modulate turn angle α. A $speed_function$ takes the current ∇s and *gain* and returns the *speed* for that time-step. Several $speed_function$s were experimented with: `constant` always returns 1mm (no stops); `random` randomly returns 0 with probability $\mathbb{P}$, and 1mm otherwise (random stops); `piecewise` (Figure 5) returns 0 if ∇s and *gain* together result in a turn angle α of $> 37°$, and 1mm otherwise (sensory-based stops). The `piecewise` function was chosen to investigate the hypothesis that a high change in sensory signal triggers stopping behaviour in real larvae. The comparison of these speed functions is a primary focus of Sections 2.3 and 2.4. Other possible speed functions are considered in Section 2.4.

A final mechanism introduced to investigate stopping behaviour was the 'gain change on stop'. When this is enabled, the speed is calculated as normal using the chosen $speed_function$, but if the result is 0, the *gain* for calculating the angle modulation is temporarily doubled for

the current time-step only. This mechanism is intended to model the hypothesis that stopping in real larvae releases a physical inhibition (caused by forward movement) of the head turn, allowing the 'head cast' turn after a stop to be larger than 'weathervaning' turns during forward movement. The comparison of the agent behaviour with and without this mechanism is a primary focus of Sections 2.3 and 2.4. Gain-change functions other than doubling are considered in Section 2.4.

The full movement algorithm for the three-point agent is thus as follows:

1. Sense ∇s as before.
2. Calculate the speed for the current time-step, $speed_i$, using $speed_i = speed_function(\nabla s, gain)$.
3. If 'gain change on stop' is enabled and $speed_i == 0$, then $gain_i = gain * 2$. Otherwise, $gain_i = gain$.
4. Calculate θ_i as before, but using $gain_i$ rather than $gain$, and taking into account physical constraints (no bending over itself, no going outside the boundaries of the environment).
5. Move. Move the head point and bend point in the direction of θ_{i-1}; straighten the tail segment as described above.
6. Turn. Pivot the head point around the bend point such that the bend-head angle becomes θ_i.

2.3 Results and analysis

The experimental hypotheses are as follows:

1. Stopping behaviour improves chemotaxis performance.
2. Sensory-based stopping behaviour produces a greater improvement in chemotaxis performance compared to random stops.
3. Doubling of the gain for turns following stops ('gain change on stop') produces a greater improvement in chemotaxis performance compared to stopping alone.

In order to investigate these hypotheses, it was first necessary to determine how best to measure 'chemotaxis performance'. Several suitable behavioural metrics were identified, most of which were taken from [1], [2] and [3], but ultimately the definition of better chemotaxis performance is whatever behaviour provides the best evolutionary advantage to the animal. This is not easy to determine, and the differences in chemotaxis behaviour across different phenotypes of larvae highlights the possibility of many different possible successful strategies, and many ecological niches to be filled. In particular, larval behaviour near an order source can be categorised as 'segmented search' vs 'orbital behaviour', as shown in Figure 6, and it is not clear which strategy is 'better' for the animal.

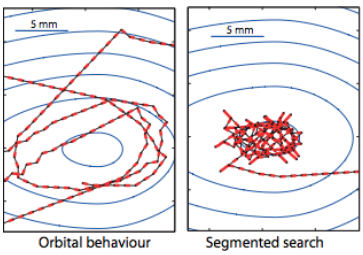

Figure 6: Near-source behaviour in single-point agent [1]

The metrics

For the purposes of this investigation, 'chemotaxis performance' of the multi-point model was determined by using behavioural metrics relating to three areas of chemotaxis behaviour. The first area of behaviour is approach behaviour. For this investigation, approach behaviour was defined as the frames from the start of a run until the first frame where the agent's midpoint becomes within 10mm of the odour source. The metrics used to describe approach behaviour were straightness index, and time to odour. The **straightness index** of run is the ratio of the straight line length of the shortest possible path between the starting point and the odour source, and the actual approach path (plus the straight line distance between the final approach position and the odour source). The straightness index lies between 0 and 1, with 1 considered the best possible value for this metric. The **time to odour** metric measures how long it takes for the agent to reach within 10mm of the odour source. This lies between 0 seconds and 'never' (a 500 second limit was used in all experiments), with lower values considered better.

The second area of behaviour is the 'far-from-source' behaviour. For this investigation, far-from-source behaviour was defined as the frames where the agent's midpoint lies outwith 10mm of the odour source. This is similar to 'approach behaviour', but can include frames where the agent has reached the odour source previously but subsequently moves further away again. The final area of behaviour is the 'near-source' behaviour. For this investigation, near-source behaviour was defined as the frames where the agent's midpoint lies within 10mm of the odour source.

The metrics used to describe and compare far-from-source and near-source behaviour were bearing-to-odour distribution, proportion of number of stops in a row, and probability of a large turn. The **bearing to odour distribution** plots the bearing of the agent to the odour (range $[-180°, 180°]$, $0°$ means facing the odour source), against the overall time spent at that bearing. Far-from-source bearing-to-odour distributions were considered better when the bearing-to-odour was near $0°$ most of the time, meaning the agent is facing the odour-source whenever it is far away. Near-to-source bearing-to-odour distributions were considered 'segmented' when the distribution is approximately equal over its range, and 'orbital' when the distribution has peaks at $-90°$ and $90°$, and troughs at $0°$ and $\pm 180°$. The **proportion of number of stops in a row** describes how often the agent stops, and whether these stops are followed by normal movement or additional stops. It was considered better when the frequency of stops was low for far-from-source behaviour and high for near-source behaviour. The **probability of a large turn** is plotted against the bearing to odour, with better behaviour far-from-source when there is a low probability of stopping when facing the odour, and a larger probability the further the bearing-to-odour is from $0°$. Better behaviour near-source would be a higher probability of large turns in general. A large turn was defined as a turn of more than $37°$, not followed by a turn of more than $37°$ in the opposite direction.

A final metric was used that describe chemotaxis behaviour across the whole run of an agent. The **distance-time** metric plots the distance from the agent midpoint to the odour centre against time. This shows the evolution of the agent's path, and for normal model parameters, it is easy to distinguish the approach and near-source phases (a steep decline from the initial distance to a stable orbiting distance).

Experimental procedure

Throughout all experiments, the following procedure was adopted:

- The three-point agent, as described in Section 2.2, was used in all experiments.
- The baseline turn angle, α_b, *gain*, *stopping_function* and *gain_change_on_stop* (GCOS) parameters were varied, as described for the particular experiment.
- The number of runs, n, was 100 for all experiments. For all graphs, one or more of the mean, median, range, interquartile range and standard deviation are shown, as deemed appropriate for each metric.
- The initial agent position for each of the n runs was varied, with a random position chosen between 40mm and 50mm from the odour source. An example such set of n initial positions is shown by Figure 7, but note that these initial positions were different for each experiment. The initial agent heading for each run was chosen randomly from the full range $[0°, 360°]$.

In addition to `constant`, `random` and `piecewise`, a further speed function was investigated - `piecewise with noise`. The noise parameter refers to Gaussian noise introduced to vary (each time step) the boundary at which a stop will occur. The boundary is centred on 37°.

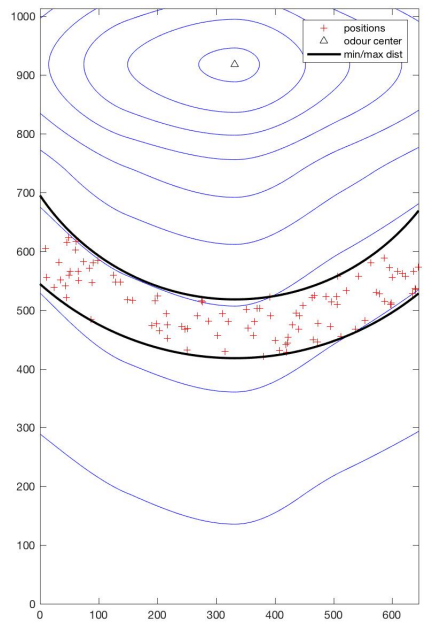

Figure 7: Example set of initial agent positions for an experiment

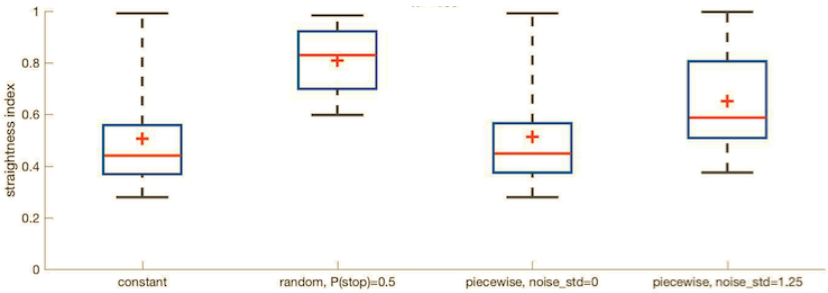

Figure 8: Straightness index, $gain = -30$, $\alpha_b = 10$, no GCOS

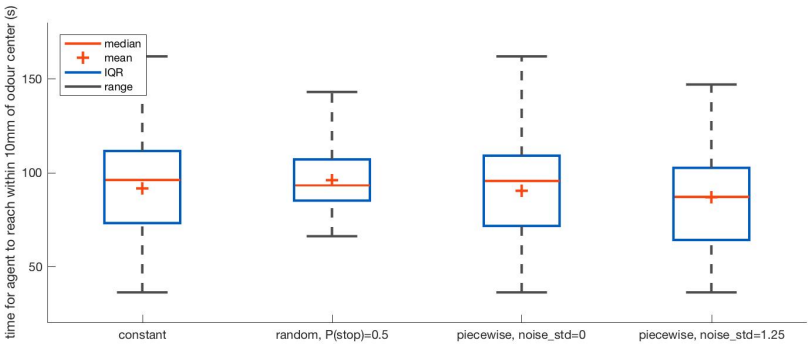

Figure 9: Time to odour, $gain = -30$, $\alpha_b = 10$, no GCOS

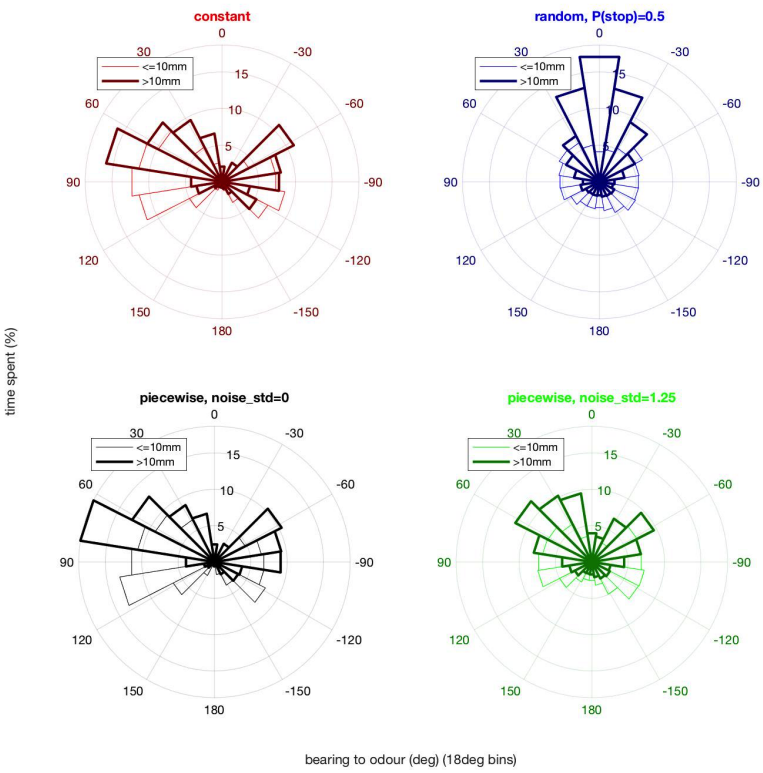

Figure 10: Bearing to odour distribution, $gain = -30$, $\alpha_b = 10$, no GCOS

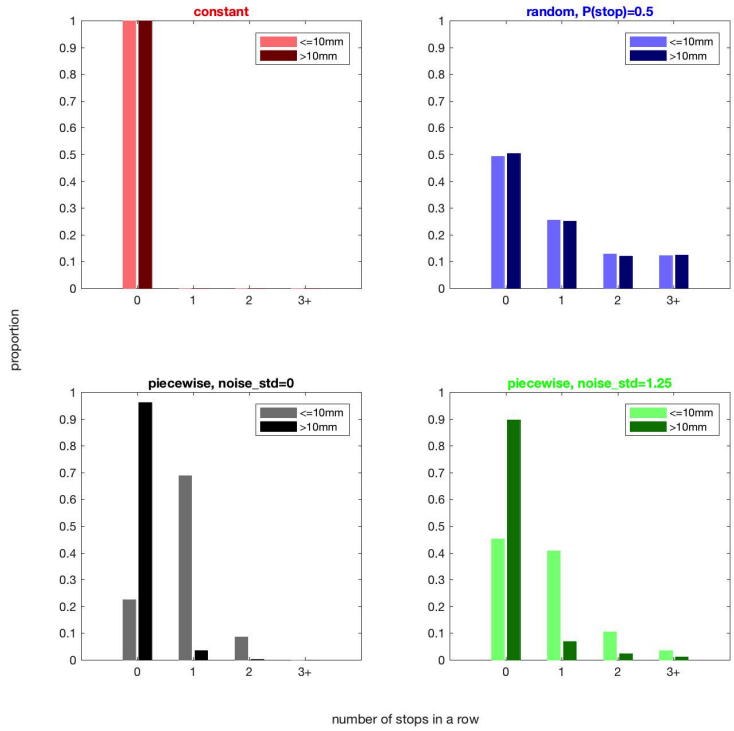

Figure 11: Proportion of number of stops in a row, $gain = -100$, $\alpha_b = 10$, no GCOS [5]

Effect of stops on chemotaxis performance

This section presents the evidence for and against hypothesis 1 - 'stopping behaviour improves chemotaxis performance'.

The most consistently obvious result throughout the experimentation phase was that the more frequent the stops on approach, the straighter the approach path. This is shown by comparing the straightness index (Figure 8) to the proportion of number of stops (Figure 11, $> 10mm$ condition) across the speed functions. The `random` speed function has the highest proportion of stops on approach, and has the highest straightness values. Indeed it was found that with a sufficiently high *gain* value, the higher the probability of stopping ($\mathbb{P}$), the higher the straightness index, with no apparent limit until $\mathbb{P}$ is high enough that the agent doesn't reach the odour source in the 500 seconds allotted. These results persist across gain values ($gain = -30, -50, -100$). Note that these results appear less significant when the baseline angle is higher (eg. 30°). This is because increasing the baseline angle has a separate straightening effect, which appears to dilute the effects of stopping.

These results are supported by the bearing to odour distribution on approach (Figure 10, $> 10mm$ condition) and distance-time on approach (Figure 13, descent from initial distance before distance stabilises). The bearing to odour distribution across speed functions shows that the more frequent the stops on approach, the more the approach is 'segmented search' (straight) compared to 'orbital' (curved). The distance-time metric shows that the descent is steeper for speed functions with stops, and the stable distance is therefore achieved sooner.

[5] Qualitative results for the $> 10mm$ condition persist across gain values ($gain = -30, -50-, 100$).

Although the previous straightness metrics correlate with chemotaxis performance in general, when stopping is involved, there seems to be a trade-off between straightness and time to odour. This is because the more stops there are on approach, the longer it takes for the agent to arrive at the odour source. Figure 9 shows that because of this trade-off, stopping frequency does not correlate exactly with time-to-odour (this is discussed further for hypothesis 2).

The above deals with approach and far-from-source behaviour. However, frequency of stops also appears to have an impact on near-source behaviour. Figure 12, $<= 10mm$ condition, shows the probability of large turns when near the odour source. For speed functions with stops, the probability of large turns at lower bearing-to-odour values increases (on balance, the probability at higher bearings decreases, but less significantly). This will cause the agent to turn more when near the odour source, a positive chemotaxis performance factor. This result is discussed further for hypothesis 2. The distance-time metric when near the odour source (Figure 13, stable distance), shows that the speed functions with stops produce a lower stable distance (closer to 'segmented search'). This distance-time result persists across gain values ($gain = -30, -50, -100$).

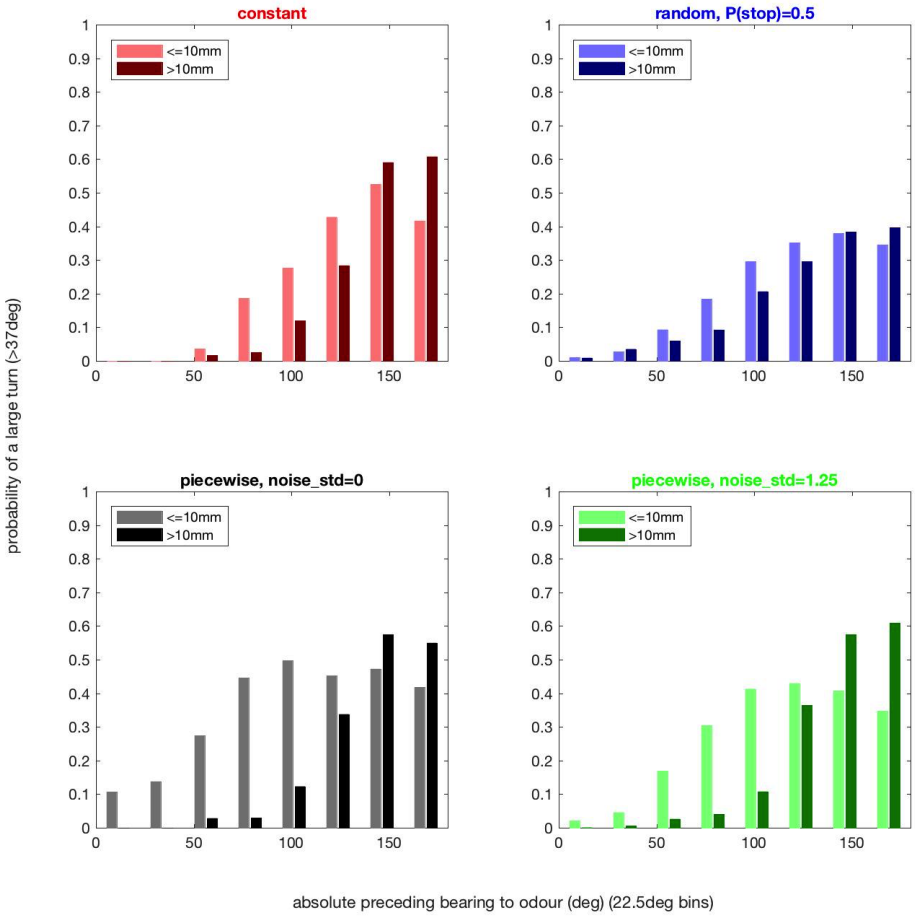

Figure 12: Probability of a large turn, $gain = -100$, $\alpha_b = 10$, no GCOS

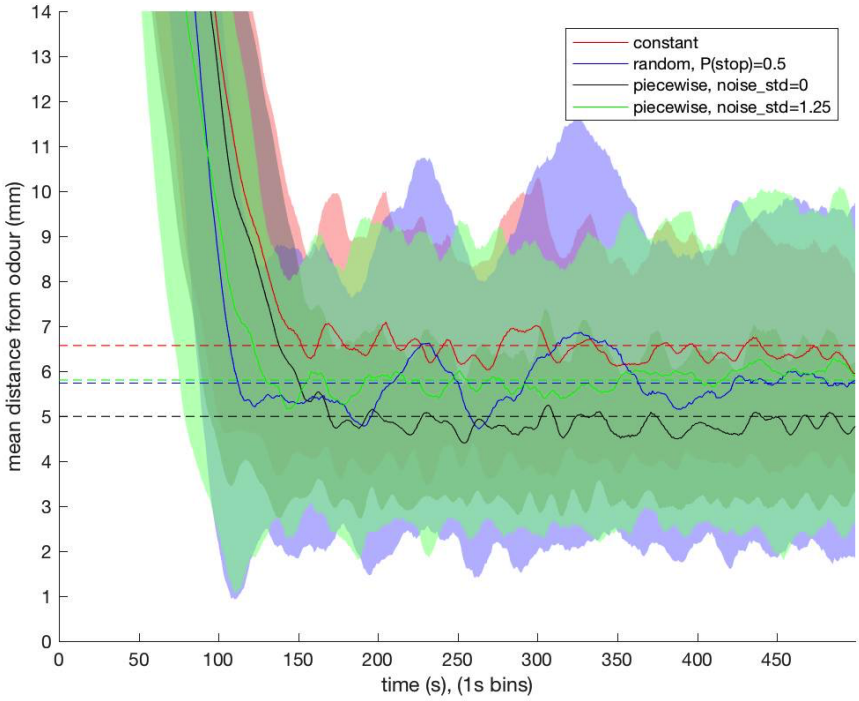

Figure 13: Mean distance-time, $gain = -30$, $\alpha_b = 10$, no GCOS[6]

Sensory-based stopping behaviour vs random stops

This section presents the evidence for and against hypothesis 2 - 'sensory-based stopping behaviour produces a greater improvement in chemotaxis performance compared to random stops'.

It has already been shown that stopping more frequently produces straighter paths compared to less frequent sensory-based stops. However, further examination of Figure 9 reveals that noisy sensory-based stops have the edge over random stops when it comes to the time-odour metric. This result persists across gain values ($gain = -30, -50, -100$). Examining the other metrics available, it is not clear why this is the case, and it might be that some quality not captured by these other metrics is giving an advantage. The simplest reason with the information available is that `piecewise with noise` happens to produce the optimal probability of stopping (approximately 0.1-0.25) out of all the speed functions experimented with. Whatever the reason, time to odour is a better indicator of chemotaxis performance compared to straightness, and so this small advantage is more significant overall.

As discussed for hypothesis 1, Figure 12, $<= 10mm$ condition, shows that there is a greater probability of large turns near the odour source for speed functions with stops compared to speed functions without stops. Closer examination reveals that this effect is more significant for sensory-based stops compared to random stops, and more significant for `piecewise` compared to `piecewise with noise`, particularly for low values of absolute bearing-to-odour.

[6]The shaded regions show the standard deviation and the dotted lines approximate the stable distance using the mean value within the interquartile range.

This suggest that for the `piecewise` speed function, the agent is likely to stay closer to the odour source. Indeed, this is confirmed by Figure 13, which shows that `piecewise` has the lowest stable distance of all the speed functions.

Gain change on stop

Little support was found **in general** for hypothesis 3 - 'doubling of the gain for turns following stops ('gain change on stop') produces a greater improvement in chemotaxis performance compared to stopping alone'. Although no evidence is given (so as to make room for the more significant results above), it was found that across the speed functions investigated, GCOS had an inconsistent effect on chemotaxis performance. Preliminary evidence suggests however, that GCOS may be more beneficial to sensory-based stops compared to random stops. Limited time was available to pursue this alternative hypothesis, and so this was left for future work.

2.4 Discussion and conclusions

The following conclusions can be drawn from the above results and analysis:

- Stopping behaviour directly affects path straightness - the more frequent the stops, the straighter the path.
- Stopping behaviour directly affects the time it takes for the agent to reach the odour source - the more frequent the stops, the longer it takes.
- Preliminary evidence suggests that the stopping probability with the best trade-off between these factors (in terms of optimising time-to-odour) is approximately 0.1-0.25 - further investigation is required.
- Preliminary evidence suggests sensory-based stops with noise produce better approach performance compared to random stops - further investigation is required.
- Stopping behaviour directly affects the ability to achieve closer 'stable distances' or 'orbits' to the odour source - stops reduce this distance in general, and in particular sensory-triggered stops produce the closest distance out of all the stopping behaviours investigated.
- Preliminary evidence suggests that doubling the gain for turns after stops increases chemotaxis performance for sensory-based stops, but not stops in general - further investigation is required.

In reference to the project aims, aims 1 and 2 have been clearly met - the model has been extended from a single point to multiple points, and extended to produce larval stopping behaviour. Aim 3 has been partially met - suitable behavioural metrics have been chosen and code written to systematically produce these metrics from the model whilst varying parameters relevant to the investigation of stopping behaviour. Such experiments have been performed and superficially analysed, but many results have been inconclusive and experimental hypotheses 2 and 3 have not been fully supported or refuted. This has resulted in aim 4 being met only minimally, with true insights into the effects of larval stopping behaviour on chemotaxis out of reach.

Several additional avenues of investigation were left unexplored due to the time constraints of this project. Suggestions for future work are included here:

Comparison to larval data The scientific value of this kind of modelling comes from comparing the behaviour of the model to the behaviour of the real-life organism. The suggestions here is to compare the behaviour of the three-point SMT model to the behaviour of real larvae, to see if this investigation's results about stopping behaviour can be applied. A good starting point might be to attempt tuning the parameters of the model (baseline angle, gain, speed function, gain change on stop) so that the relevant metrics match as closely as possible to those of larval data from [2] and [3].

Investigate optimal stopping probability An approximate range of values was suggested, but future work might systematically investigate this question, and produce a prediction which can be tested against real larval behaviour.

More complex speed functions All speed functions used in this investigation can produce only two possible speeds (0 or $1mm/s$). Missing from this is the idea of a maximum speed, achievable only when the agent is moving in a straight line. A more interesting function would have a continuous range of values from the maximum speed all the way down to 0, with a base speed that is modulated by the sensory signal (similar to how turn angle is modulated presently) - the differences in paths produced by a sigmoid function compared to a linear function might present an interesting starting point. Note that these extensions are easily achieved within the current model framework.

Different gain changes on stop Although the addition of 'gain change on stop' did not produce conclusive results, it might be that having a different gain change function does - in this investigation only the doubling function was used, but really there is no limit to what function could be used. A closer examination of the peristaltic inhibition in larvae may produce the best starting point for such an investigation.

Different odour maps Only one odour map was used throughout this investigation, based on odour data from a particular live experiment. Before drawing any conclusions about chemotaxis in general, it would be good idea to confirm that the results of this investigation still hold when different odour maps are used, either artificial (Gaussian, exponential, shallow linear, steep linear), or from additional odour samples.

Moving bend point The bend point in the multi-point agent is set at the start and doesn't change throughout a run. However, observing real larval movement reveals that for small turns, the bend point is located nearer the head of the agent, whereas for large turns the bend point is nearer the midpoint. Whether or not this is worth investigating needs to be considered further - it might be that the midpoint bend point approximation captures the freedom of movement of the larvae sufficiently.

Continous time model The SMT vs STM dilemma (see Appendix A) highlights how the discrete approximation of time used in the model affects how the agent moves. This is undesirable if the goal is to compare the model agent's movement to larval movement. For this reason, it would be worth attempting to confirm the results of this investigation in a continuous model where, as in real larvae, the move rhythm is separated from the turn rhythm.

When does sensing occur For a truly accurate movement approximation, the above suggestion would first have to answer the question of when sensing occurs in real larvae. Is it attached to the move cycle, turn cycle, does sensing have its own rhythm, or is sensing actually continuous?

2.5 Acknowledgements

I would like to thank my supervisor Barbara Webb for encouraging me to apply for this scholarship, and for providing vital insight and direction throughout the project.
I would also like to thank Barbara Webb, Konstantinos Lagogiannis and Antoine Wystrach for giving me early access to their paper [1], for coming up with the ideas and motivation for this project, and for answering all of my questions relating to their research.
Finally I would like to thank the people of the Minimal research group at the School of Informatics for warmly including me in their weekly meetings.

References

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A STM vs SMT

Figure 14 shows the differences in movement between the single-point agent, three-point agent with STM, and three-point agent with SMT, all other factors staying the same. Each of the agents start with their head point at [0, 0], facing 45° anticlockwise from east. Each of the agents receive the same sensory signal, and so the next heading θ_i is computed to be the same value for all of them, and is assumed to be 135° (a left turn of 90°). Note that the length of the lines in Figure 14 are important to the results - the length of the head segments in the three-point agents is 2mm, whereas all forward movements are 1mm in distance.

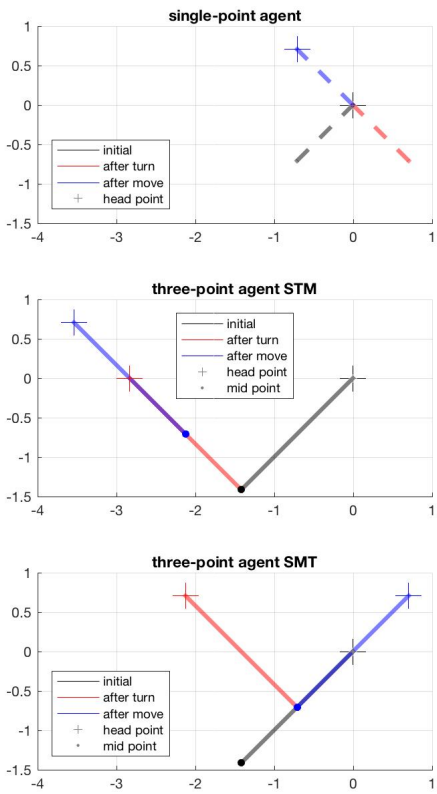

Figure 14: Differences in movement styles (all units in mm).

In the single-point and STM agents, the turn step occurs first, and so θ_i is adopted before moving. After the turn, the forward movement is the same for both. The reason for the dramatic difference in where the final head point ends up is that the single-point agent pivots on itself, whereas the STM agent pivots from its bend point[7].

For the SMT agent, the move step occurs first, in the direction of θ_{i-1}. Whilst this is different from the single-point agent's forward movement, it actually alleviates some of the displacement of the final head point caused by the following turn movement, resulting in movement behaviour slightly closer to that of the single-point agent[8].

This analysis reveals fundamental differences between these three movement styles. In terms of modelling larval behaviour as accurately as possible, none of these movement styles reflect exactly how the larvae move - [1] show that the larvae actually have separate rhythms for forward (peristaltic) movement and turning movements (lateral oscillation). This makes the choice between STM and SMT arbitrary in terms of realism. For the purposes of this investigation however, a decision had to be be made, and it seemed preferable to choose the movement style that produces behaviour closest to that of the single-point agent, simply so that easier comparisons could be drawn during the preliminary investigations into stopping behaviour. For this reason, the choice was made to use the SMT movement style throughout this investigation.

B Physical constraints

The bend angle at each time step is defined as the angle between the head and tail segments. In order to prevent the multi-point agent from bending over itself or bending itself more than physically possible, the turn angle α at each time-step is constrained such that the resulting

[7]The closer the bend point is to the head point, the closer the STM agent's path is to that of the single-point agent, all other factors staying the same.

[8]The closer the speed is to the length of the head segment, the closer the the SMT agent's path is to that of the single-point agent, all other factors staying the same.

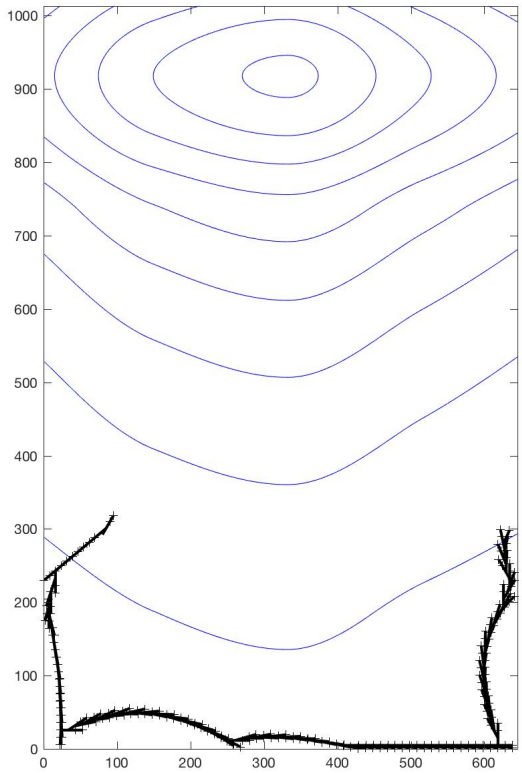

Figure 15: Aversion behaviour ($gain = 50$), demonstrating collision prevention.

turn does not cause the resulting bend angle to become less than some value, chosen to be 20° throughout this investigation. This is achieved each time-step by calculating the angle between the head and tail segments in the current *oscillation_direction*, and forcing the calculated turn angle α to be at most equal to this value minus 20°.

If not prevented, the agent can travel outside the boundaries of its environment, causing an error during the sensing stage when there is no sense value to provide for the location of the head point. To solve this issue, the single-point and multiple-point agents were extended to predict if the head point is going to cross the boundary and modify their course to prevent this. This is achieved each time-step by projecting forward where the head point is going to end up, and setting the current heading to be parallel to the boundary if a collision would otherwise occur. The direction parallel to the boundary is chosen to be as close to the agent's original direction as possible, unless this would result in the agent getting stuck in a corner. For the multi-point agent, in the case where a move forward action will cause a collision, the agent's *speed* for that step is set to 0.

The result of this algorithm is that the agent will move parallel to any boundary it collides with until it decides to move away from the boundary due to some stimulus. This is roughly similar to what real larvae do. An example is shown in Figure 15. The video of this example path can be accessed from[9]. Note that collision prevention overrides the bending constraints described above.

[9]aversion behaviour: `https://youtu.be/Im1CHp4pQmY`